

Disparities by Race/Ethnicity and Sex in the Relationship Between Adverse Childhood Experiences (ACEs) and Suicidality

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Does not have any relevant disclosures

Background

- Adverse childhood experiences (ACEs) are traumatic events that happen from ages 0-17 years old are risk factors for suicidal ideation (SI) and attempt (SA).
- There are racial/ethnic and sex disparities on the types of ACEs experienced.
- Incorporating distinct groups of ideators and attempters into research analysis is needed.
- Advanced causal inference and data science approaches offer the ability to simulate randomized control trials (RCTs) in observational studies.
- Causal estimates in a multiple exposure setting, along with incorporating complex survey design, is both limited in suicide research and other areas.

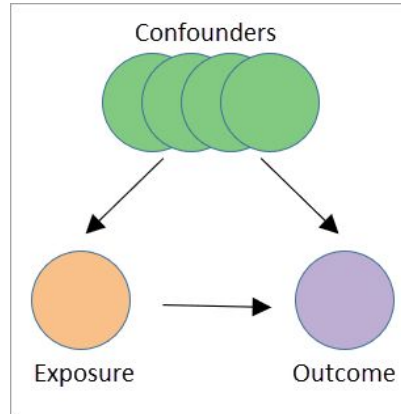
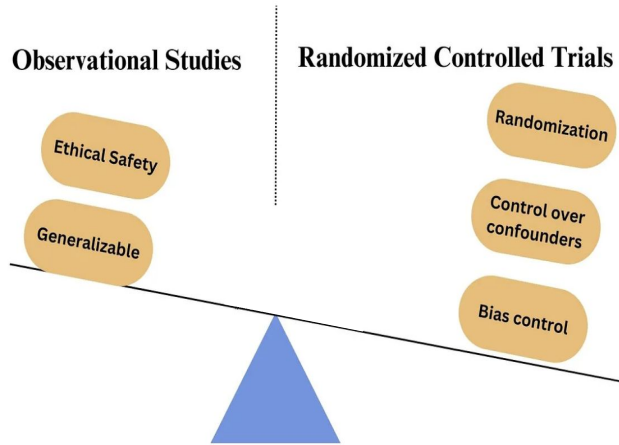
Motivation/Purpose

- Given the abundance of cross-sectional datasets in public health, we wanted to build a generalizable causal workflow on a cross-sectional dataset that can then be used on a longitudinal dataset.
- Provide robust estimates of a dose response relationship between ACEs and suicidal ideation and attempt.
- Test for disparities in the dose response relationship in the demographics (race/ethnicity and sex).

Methods

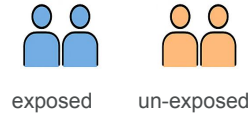
- **Sample:** Youth (n = 20,103) from the YRBS 2023*
 - **Outcomes:** 11.7% of youth endorsed suicidal ideation only (SI) and 9.5% endorsed having a suicide attempt (SA).
 - **Exposure:** 23.6% didn't have ACEs, 24.9% had one ACE, 19.2% had two ACEs, 13.8% had 3 ACEs, and 18.5% had 4 or more ACEs.
 - **Demographics:** 48.1% White, 27.4% Hispanic, 13.3% Black, 11.1% All Other Races (e.g., Asian, Multiracial), 51.9% were males, and a similar distribution across grades 9 to 12 (23.4% to 26.5%).
 - **Confounders:** Safety concern at school, experienced racism, bullied in school, bullied online, feeling sad or hopeless, ever vaped, ever smoked a cigarette, first alcoholic drink before the age of 13, ever smoked weed, ever use of heroin, methamphetamine or misuse of prescription drugs, and being physically active (more than 5 days a week for 60 minutes, attending gym class or being on a sports team)

Randomized Control Trial vs Observational Studies



Exchangeability in Observational Data

Before Adjustment
(Not Exchangeable)

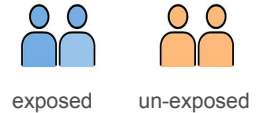


Young → Old

Groups differ due to age
→ confounding.

Cannot estimate true effect

After Adjustment for Age
(Exchangeable)



After controlling for age,
groups are exchangeable

True effect of exercise
can be estimated

Causal Inference Approaches

- **Inverse Probability of Treatment Weighting (IPTW):** A causal inference method which makes exposure groups (e.g., having ACEs vs not having ACEs) exchangeable on their other potentially confounding characteristics (e.g., race/ethnicity or substance use history).
- **Weighted Logistic Regression:** Using both survey and IPTW weights, we can use a weighted logistic regression to predict an outcome (e.g., suicide ideation) based on the exposure (e.g., ACE) for an unbiased estimate.
- **G-Computation:** A causal inference method used with observational data to estimate the effects of exposures on outcomes by creating hypothetical scenarios (potential outcomes) for individuals under alternative exposure patterns and then comparing them.

Methods

- **Statistical Analysis**

- **ACE Prevalences:**

- Display the prevalences of the ACE types by race/ethnicity and sex with using the Pearson-Chi Square test with Rao and Scott adjustment

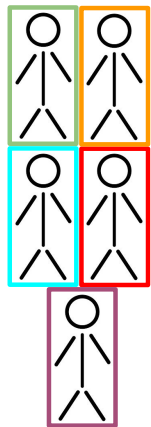
- **Dose-Response Relationship between ACEs and SI/SA using Doubly Robust Estimation:**

- **Step 1:** We predict ACEs with sex, race/ethnicity, primary sampling unit and stratum as predictors (all effecting ACEs) with a Xgboost model to get the propensity score then compute the inverse probability of treatment weighting and combine with the survey weight.
 - **Step 2:** Using a weighted logistic regression model we predict SI and SA with demographics and confounders with accounting for the survey design.
 - **Step 3:** After fitting the weighted logistic regression model, we apply a G-computation (counterfactual predictions) and then compute adjusted risk ratios (aRR) overall, by race/ethnicity and sex.

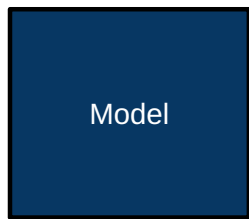
- **Dose-Response Disparities:** We test for moderation by race/ethnicity and sex between ACEs using an ANOVA test.

Doubly Robust Estimation Framework

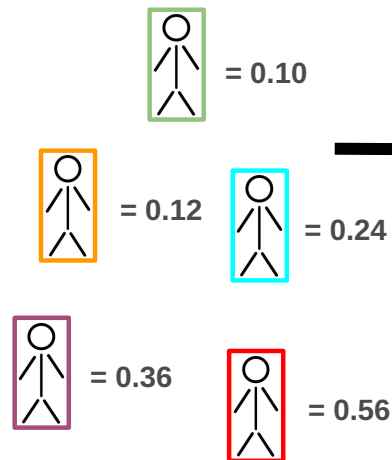
Exposure Groups
After IPTW



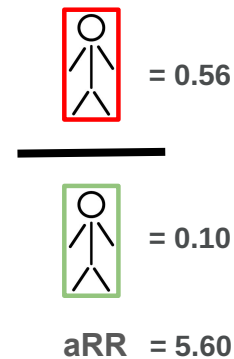
Build Model to
Predict SI and SA



Apply
G-Computation



Compute
Adjusted Risk Ratios



Results: ACE Prevalences by Race/Ethnicity

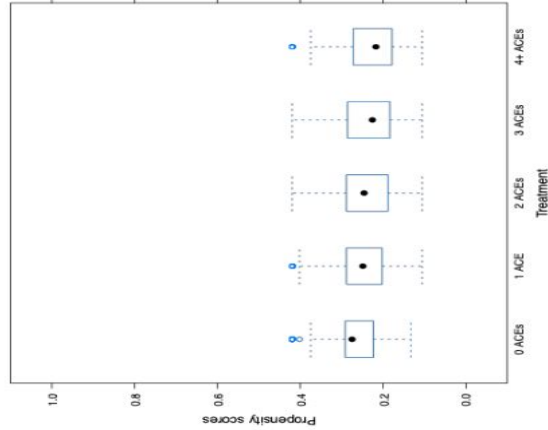
White	All Other Races	Black or African American	Hispanic/Latino
Household substance use	Emotional abuse	Parent or guardian incarcerated or detained	Physical neglect
Household poor mental health	Witnessed intimate partner violence	Physical abuse	Sexual abuse

Results: ACE Prevalences by Sex

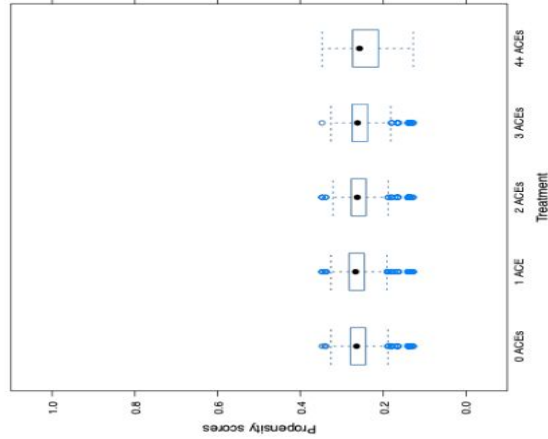
Males	Females
Physical neglect	Emotional abuse
	Parent or guardian incarcerated or detained
	Witnessed intimate partner violence
	Household substance use
	Household poor mental health
	Physical abuse
	Sexual abuse

Results: Overlap

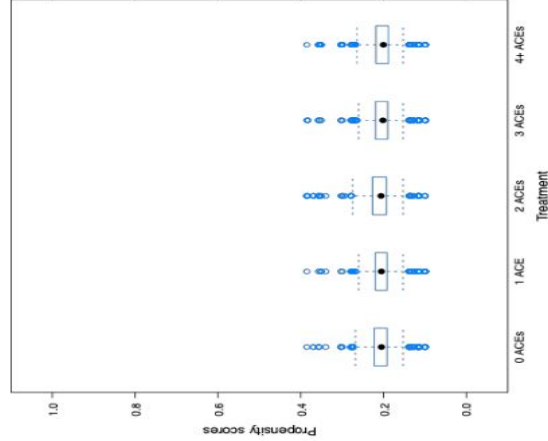
0 ACEs propensity scores by Tx group



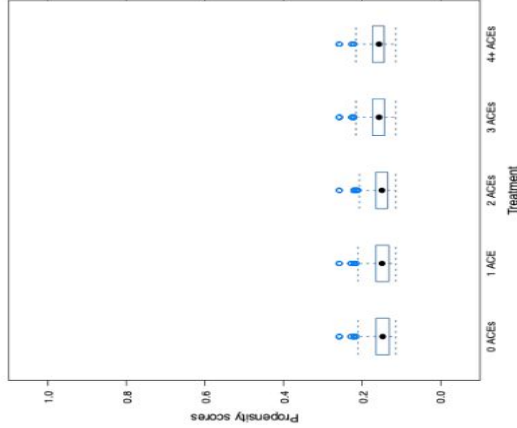
1 ACE propensity scores by Tx group



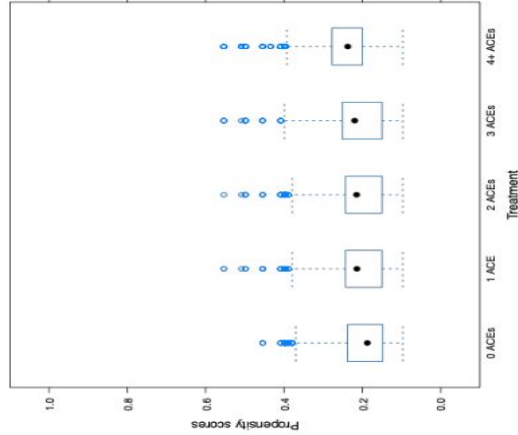
2 ACEs propensity scores by Tx group



3 ACEs propensity scores by Tx group



4+ ACEs propensity scores by Tx group



Results: Balance

Variable	Maximum Standardized Mean Difference
Sex	0.10
Race/ethnicity	0.05
Stratum	0.04
Primary Sampling Unit	0.06

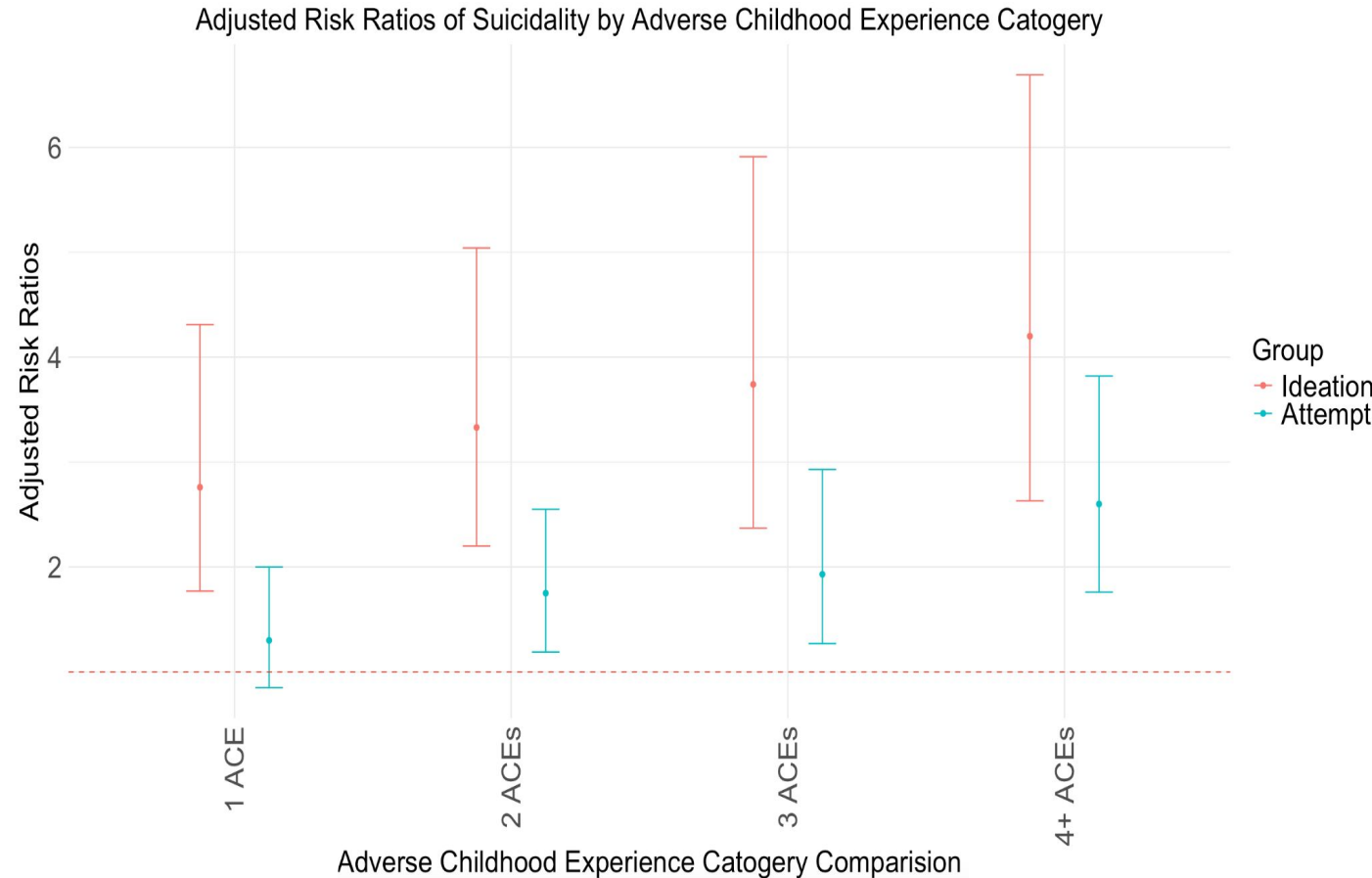
Results: AUCROC and AUCPRC - Propensity Model

Outcome	AUCROC	AUCPRC
0 ACEs	0.61 (0.60,0.62)	0.30 (0.30,0.31)
1 ACEs	0.55 (0.54,0.56)	0.27 (0.26,0.28)
2 ACEs	0.54 (0.53,0.55)	0.21 (0.21,0.22)
3 ACES	0.55 (0.54,0.56)	0.17 (0.16,0.17)
4 + ACEs	0.64 (0.63,0.64)	0.34 (0.33,0.35)

AUCROC: Area under the curve receiver-operating characteristic curve;
AUCPRC: Area under the curve precision recall curve

Results: Dose-Response Effect of ACEs on SI/SA

Finding: A dose response relationship between ACEs and suicidality was observed for both SI and SA



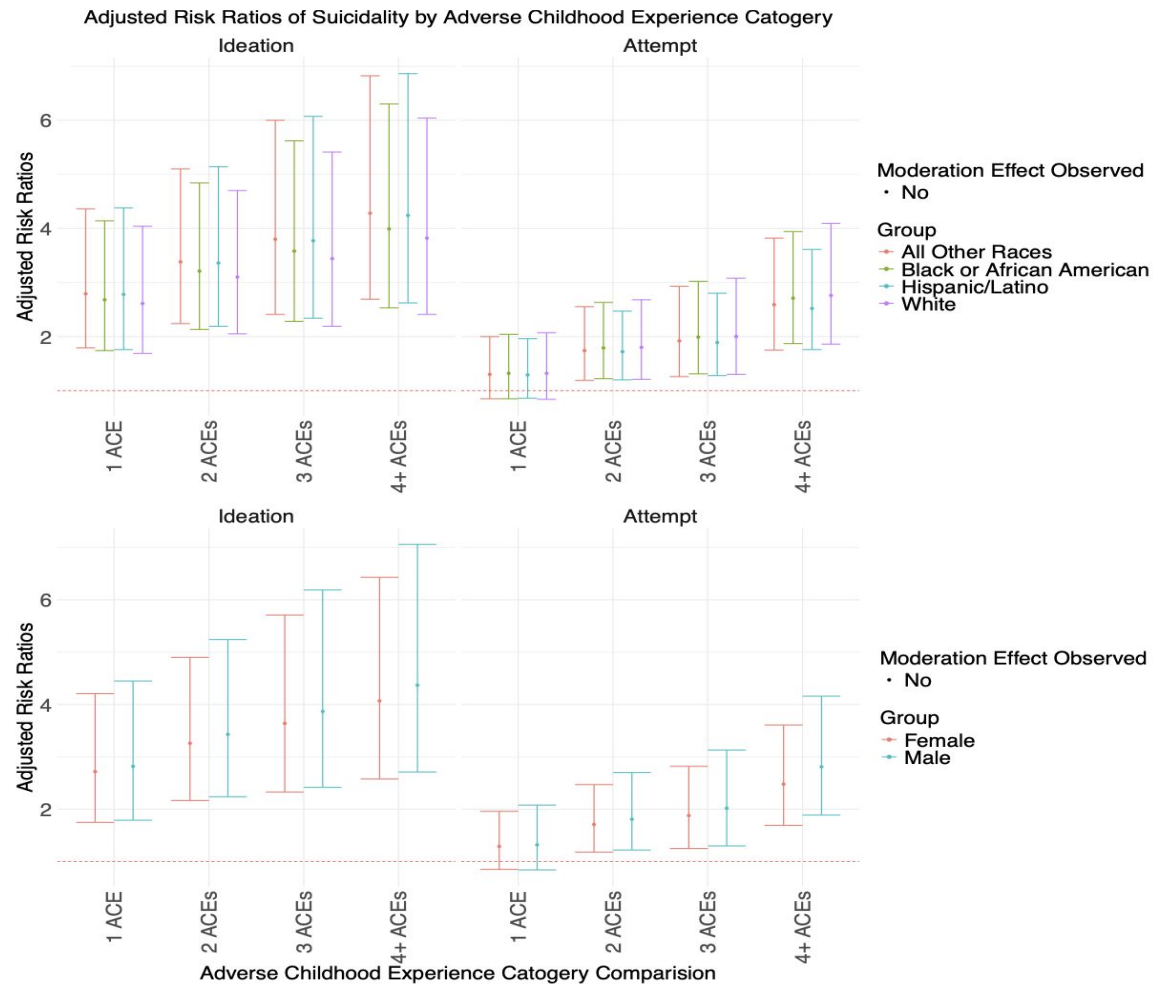
Results: AUCROC and AUCPRC - Outcome Model

Outcome	Metric	Outcome Model Metrics	Outcome Model on Counterfactual Grid of ACEs*
Ideation	AUCROC	0.79 (0.78,0.80)	0.75 (0.75,0.75)
Ideation	AUCPRC	0.31 (0.30,0.31)	0.27 (0.27,0.28)
Attempt	AUCROC	0.87 (0.86,0.87)	0.84 (0.84,0.85)
Attempt	AUCPRC	0.44 (0.43,0.45)	0.39 (0.38,0.39)

*Predictions were computed without using a R package;
AUCROC: Area under the curve receiver-operating characteristic curve;
AUCPRC: Area under the curve precision recall curve

Results: Dose-Response Disparities by Race/Ethnicity and Sex

Finding: A dose-response relationship between ACEs and suicidality is maintained across races/ethnicities and sexes with no evidence of disparities.



Discussion

- In a nationally representative sample, we estimated the effect of ACEs on suicidality in adolescents by demographics using advanced methods in a causal inference framework with accounting for the survey design.
- Overall and within demographics, having 1 or more ACEs for suicide ideation or 2 or more ACEs for attempt put youth at risk for these outcomes despite the difference in prevalences at the demographic level.
- These results suggest that we can culturally tailor interventions to youth experiencing ACEs in suicide ideation and attempt due to their distinct characteristics.

Limitations

- We can't make causal claims in the analysis.
- The retrospective assessment of ACEs also comes with the possibility of selection bias such that the sample will only include children whose ACEs were not life threatening.
- We believe there are other factors that may affect ACEs, suicide ideation and attempt that were not in the YRBS dataset (e.g., Neighborhood Deprivation Index) that also may lead to better model metrics.

Future Directions & Impact

- Given disparities in ACE prevalences, more research is needed to understand culturally responsive strategies for preventing ACEs and understanding disparities in risk and protective factors.
- The results evidenced in this study suggest that preventing ACE's might have a greater impact on SI than SA, but future work leveraging longitudinal data should determine these mechanistic relationships.
- Continuing the reproducibility, causal machine learning (e.g., sample splitting), and reporting predictive performance on the models.

Paper, Code and Resources

- **Under Review:** Journal of the American Academy of Child and Adolescent Psychiatry (JAACAP)
- **Slides/Code:** https://github.com/adrianharrisAIhub/JAACAP_ACEs
- **Email:** ah5588@nyu.edu

SCAN ME



Appendix

Causal Inference Assumptions

- **Ignorability:** No unmeasured confounding so we can compare the exposure groups on some outcome.
- **Overlap:** People have a non-zero probability of being in the exposed or unexposed group.
- **Consistency:** Given the exposure, is what we observed for their outcome... what we would expect for that person?
- **No interference:** A person in an exposure group doesn't effect the outcome for a person in the unexposed group (and vice versa).

Results: ACE Prevalences by Race/Ethnicity

Characteristic	White N = 9,700	All Other Races N = 4,248	Black or African American N = 1,791	Hispanic/Latino N = 3,994	P-Value
Physical neglect, No. (%)	544 (7.5%) (5.9%, 9.5%)	290 (7.0%) (4.9%, 9.8%)	187 (12.4%) (9.8%, 16%)	414 (12.9%) (10.0%, 16.0%)	<0.001
Emotional abuse, No. (%)	4,034 (65.3%) (62.0%, 68.0%)	1,874 (66.1%) (61%, 71%)	846 (54.4%) (48.0%, 61.0%)	1,675 (57.0%) (54.0%, 60.0%)	<0.001
Parent or guardian incarcerated or detained, No. (%)	911 (12.7%) (10.0%, 16.0%)	645 (14.1%) (9.7%, 20.0%)	274 (17.0%) (14.0%, 20.0%)	503 (16.1%) (14.0%, 19.0%)	0.033
Witnessed intimate partner violence, No. (%)	1,165 (16.9%) (15.0%, 19.0%)	831 (23.1%) (21.0%, 26.0%)	319 (19.2%) (16.0%, 22.0%)	668 (19.6%) (17.0%, 22.0%)	0.004
Household substance use, No. (%)	1,818 (28.4%) (25.0%, 32.0%)	956 (23.3%) (19.0%, 29.0%)	257 (16.4%) (15.0%, 18.0%)	774 (24.5%) (22.0%, 27.0%)	<0.001
Household poor mental health, No. (%)	2,178 (33.0%) (30.0%, 36.0%)	927 (28.9%) (24.0%, 34.0%)	316 (19.3%) (17.0%, 22.0%)	858 (27.0%) (24.0%, 30.0%)	<0.001
Physical abuse, No. (%)	1,768 (27.4%) (25.0%, 30.0%)	1,073 (37.4%) (34.0%, 41.0%)	612 (37.9%) (31.0%, 46.0%)	1,003 (33.5%) (31.0%, 37.0%)	<0.001
Sexual abuse No. (%)	465 (6.4%) (5.1%, 8.2%)	278 (6.8%) (5.4%, 8.5%)	118 (6.2%) (4.8%, 8.0%)	303 (9.1%) (8.2%, 10%)	0.021

Results: ACE Prevalences by Sex

Characteristic	Male N = 10,061	Female N = 9,884	P-Value
Physical neglect, No. (%)	804 (10.5%) (8.8%, 12.0%)	645 (8.7%) (6.8%, 11.0%)	0.017
Emotional abuse, No. (%)	3,760 (54.1%) (51.0%, 57.0%)	4,723 (68.8%) (66.0%, 72.0%)	<0.001
Parent or guardian incarcerated or detained, No. (%)	1,077 (13.3%) (11.0%, 16.0%)	1,274 (15.6%) (13.0%, 18.0%)	0.009
Witnessed intimate partner violence, No. (%)	1,190 (14.7%) (13.0%, 17.0%)	1,811 (23.2%) (21.0%, 25.0%)	<0.001
Household substance use, No. (%)	1,619 (20.3%) (18.0%, 23.0%)	2,188 (29.6%) (27.0%, 32.0%)	<0.001
Household poor mental health, No. (%)	1,645 (22.1%) (20.0%, 25.0%)	2,640 (35.5%) (33.0%, 38.0%)	<0.001
Physical abuse, No. (%)	2,111 (29.8%) (27.0%, 32.0%)	2,395 (34.1%) (32.0%, 36.0%)	0.003
Sexual abuse, No. (%)	241 (2.7%) (2.1%, 3.5%)	929 (12.0%) (11.0%, 13.0%)	<0.001

Results: AUCROC and AUCPRC

Base Comparison: Seen in Paper

- Propensity Model: XgBoost
- Outcome Model: Generalized Linear Model (GLM) with G-Computation using the marginal effect package
- Testing for: Does this display the same aRRs as Approach 1

Approach 1:

- Propensity Model: XgBoost
- Outcome Model: GLM with G-Computation without using the marginal effect package
- Testing for: Does this display the same model metrics (outcome model) as the base comparison so we can compare to Approach 3

Approach 2:

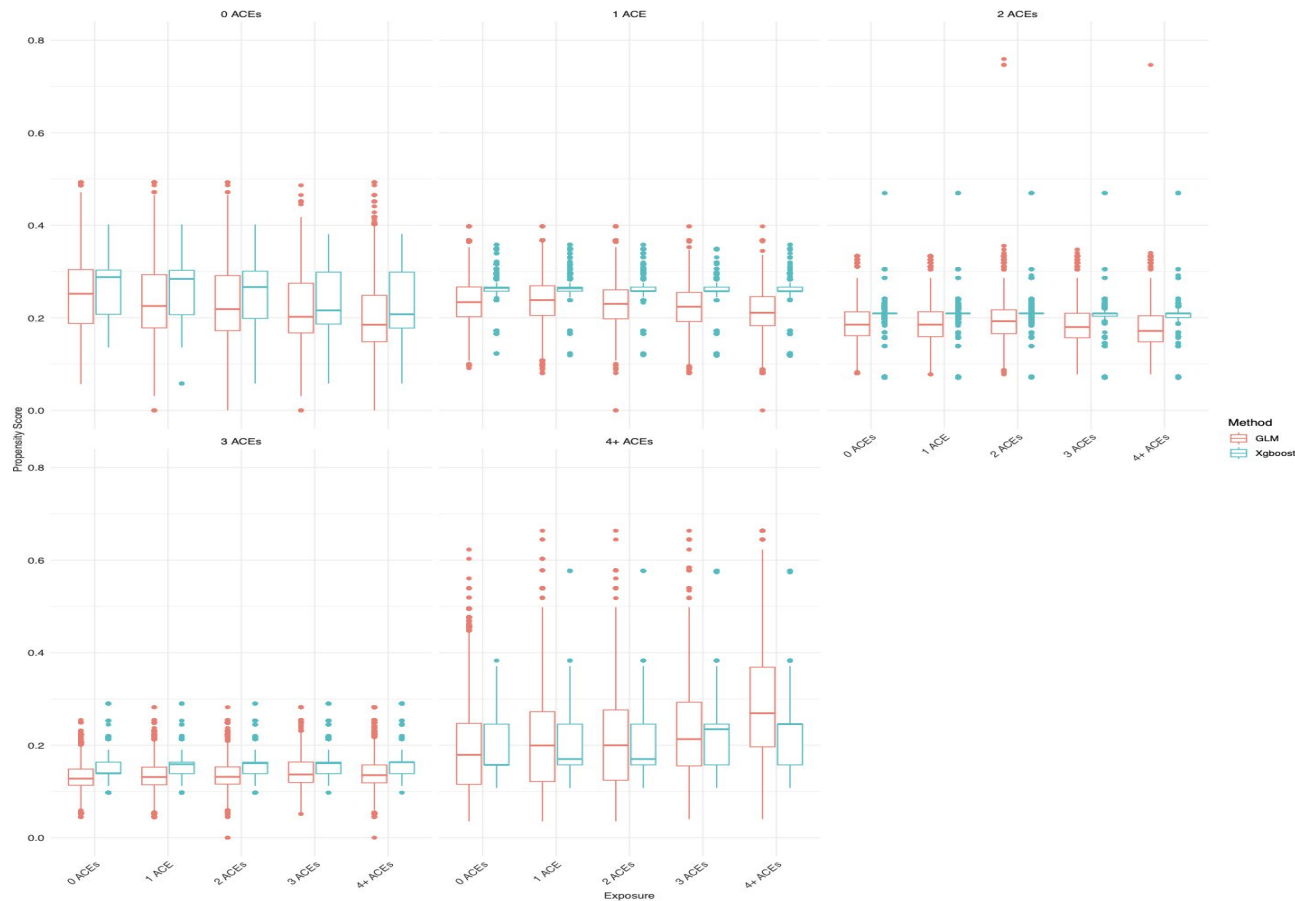
- Propensity Model: GLM
- Outcome Model: GLM with G-Computation without using the marginal effect package
- Testing for: Is this worse than Approach 1 for the model metrics (outcome model) ? If it's worse, then using “ML” is better

Approach 3:

- Propensity Model: XgBoost
- Outcome Model: XgBoost w/ G-Computation without using the marginal effect package
- Testing for: If this is better than Approach 1 and 2, on the outcome model then we can say ML is better for fitting the data on the outcome which gives more precise adjusted risk ratios

Results: Overlap - Propensity Model

- We were able to recreate the overlap plot from the twang package
- We also used a GLM as a propensity model due to the fact that XgBoost may be too complex for such few predictors



Results: AUCROC and AUCPR - Propensity Model

Exposure	AUCROC - XgBoost	AUCROC - GLM	AUCPRC - XgBoost	AUCPRC - GLM
0 ACEs	0.61 (0.60,0.62)	0.63 (0.62,0.63)	0.30 (0.30,0.31)	0.31 (0.30,0.32)
1 ACE	0.55 (0.54,0.56)	0.58 (0.57,0.59)	0.27 (0.26,0.28)	0.28 (0.27,0.28)
2 ACEs	0.54 (0.53,0.55)	0.57 (0.57,0.58)	0.21 (0.21,0.22)	0.23 (0.23,0.24)
3 ACEs	0.55 (0.54,0.56)	0.58 (0.57,0.59)	0.17 (0.16,0.17)	0.17 (0.17,0.18)
4+ ACEs	0.64 (0.63,0.64)	0.67 (0.67,0.68)	0.34 (0.33,0.35)	0.38 (0.37,0.39)

- The GLM was better on both model metrics

Results: Dose Response - Adjusted Risk Ratios

Outcome	Method	1 ACE	2 ACEs	3 ACEs	4+ ACEs
Ideation	Base	2.74 (1.70,4.41)	3.32 (2.14,5.15)	3.74 (2.31,6.06)	4.23 (2.58,6.92)
Attempt	Base	1.34 (0.86,2.09)	1.79 (1.21,2.64)	1.99 (1.31,3.04)	2.67 (1.81,3.95)
Ideation	Approach 1	2.76 (1.76,4.18)	3.32 (2.10,5.09)	3.71 (2.34,5.46)	4.16 (2.71,6.16)
Attempt	Approach 1	1.43 (0.67,2.73)	1.91 (0.92,3.47)	2.14 (1.06,4.11)	2.91 (1.45,5.42)
Ideation	Approach 2	2.69 (1.66,4.35)	3.26 (2.11,5.05)	3.62 (2.24,5.86)	4.21 (2.58,6.87)
Attempt	Approach 2	1.33 (0.86,2.07)	1.72 (1.16,2.55)	1.91 (1.25,2.91)	2.51 (1.70,3.72)
Ideation	Approach 3	1.50 (0.87,2.47)	1.77 (1.04,2.86)	2.09 (1.24,3.35)	2.28 (1.37,3.56)
Attempt	Approach 3	1.06 (0.52,2.02)	1.37 (0.67,2.49)	1.54 (0.76,2.95)	2.14 (1.04,4.04)

- Since Approach 1 is similar to the base, we hypothesize that we can compare Approach 3 to the other methods
- Approach 1 and 3 are computed with bootstrapping (n = 1,000) on a fraction of the sample

Results: AUCROC and AUCPRC - Outcome Model

Outcome	Metric	Fitted Model Approach 1	Grid Model Approach 1	Fitted Model Approach 2	Grid Model Approach 2	Fitted Model Approach 3	Grid Model Approach 3
Ideation	AUCROC	0.79 (0.78,0.80)	0.75 (0.75,0.75)	0.79 (0.78,0.80)	0.75 (0.75,0.75)	0.86 (0.85,0.87)	0.80 (0.80,0.80)
Ideation	AUCPRC	0.31 (0.30,0.31)	0.27 (0.27,0.28)	0.31 (0.30,0.31)	0.27 (0.27,0.28)	0.50 (0.49,0.51)	0.34 (0.34,0.35)
Attempt	AUCROC	0.87 (0.86,0.87)	0.84 (0.84,0.85)	0.86 (0.86,0.87)	0.84 (0.84,0.85)	0.90 (0.89,0.91)	0.86 (0.86,0.87)
Attempt	AUCPRC	0.44 (0.43,0.45)	0.39 (0.38,0.39)	0.44 (0.43,0.45)	0.39 (0.39,0.39)	0.57 (0.56,0.57)	0.45 (0.45,0.46)

AUCROC: Area under the curve receiver-operating characteristic curve;
AUCPRC: Area under the curve precision recall curve
Bold: Highest Metric